

Empirical Performance Summary

Sub-Millisecond Ingestion, Microsecond Deserialization & Streaming Throughput

The Complete Performance Scorecard

Every layer of the Fathom stack has been profiled and optimized for microsecond-level execution speed, ensuring instantaneous response times across all enterprise workflows.

MEMORY INGESTION

94 μs

PER-FACT ABSORB (100 BATCH)

5.1ms to absorb and dedup 100 facts via SHA-256 fast path.

HYBRID RETRIEVAL

1.62 ms

SEARCH MEDIAN LATENCY

Fused Vector + BM25 search across 500 facts in under 2ms.

XML FEED STREAM

1.11M

ITEMS / SEC THROUGHPUT

Streaming quick-xml parser processes 12k feed items in 10.8ms.

Parser Scaling & Extraction Latencies			
DOCUMENT TYPE	SIZE / ROWS	PARSE DURATION	EFFECTIVE THROUGHPUT
HTML Table (Scraper)	191 KB / 3,000 rows	9.55 ms	314,268 rows / sec
HTML Large (Scraper)	773 KB / 12,000 rows	34.22 ms	350,723 rows / sec
JSON Tree Walk	4 MB / 20,000 objects	43.47 ms	Stateless Parallel-Safe
AST Code Map (240 files)	3.3 MB Rust Source	34.20 ms	4,330 Summary Lines

SYSTEM LATENCY GUARANTEES

All core internal operations—from policy validation to database inserts—are strictly bounded under 5 milliseconds, ensuring that human operators and external API callers experience instantaneous execution.

Deterministic Excellence: Fathom turns AI agent execution from a sluggish, bloated prototype into a high-frequency trading caliber software runtime.